Reg. No. | | | | | | | | | | | [18MTFT202]

SECOND SEMESTER M.TECH EXAMINATION, JUNE – 2019

FOOD SAFETY AND QUALITY CONTROL

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i. Answer FIVE full questions. Question No. 1 and 2 are compulsory.**
- **ii. Draw neat sketches wherever necessary.**
- **iii. Make suitable assumptions wherever necessary.**

1.	**a.**	Recall the importance of calibrating the equipments before initiating any analysis.	**10**
	b.	Compare the sampling plans for raw materials and finished food products.	**10**
2.	**a.**	Differentiate between physically and chemically bound water in food substances. Strengthen your answer with any one methods adopted for analysis of above mentioned types of water in food.	**12**
	b.	Summarize on the decomposition reactions in food samples during the estimations of moisture content.	**08**
3.	**a.**	Under which circumstances a food analysist will prefer the atomic spectrophotometry for food safety analysis. Strengthen your answer with suitable examples.	**08**
	b.	With the help of an appropriate example, analyze the working principle of HPLC for various food samples.	**12**
		OR	
4.	**a.**	A sample of food was suspected with the presence of an unknown toxic compound. Predict the analytical method and its working principle, which will be employed by the food analyst.	**10**
	b.	Formulate the working principle of GC-MS with a neat schematic diagram of various components.	**10**
5.	**a.**	Predict the consequences of implementing a manufacturing unit without a well-defined HACCP plans.	**10**
	b.	With an appropriate food processing unit discuss good manufacturing process, which need to be considered in food industry.	**10**
		OR	
6.	**a.**	List out various physical, chemical and biological hazards and discuss the control measures in food industry	**10**
	b.	Investigate the criteria's to be considered, before applying for ISO certification.	**10**
7.	**a.**	Critically discuss on "Operations of Codex alimentarius".	**20**
		OR	
8.	**a.**	Prioritize the importance of the following in the food safety and quality control: **i.** WHO **ii.** FDA **iii.** WTO **iv.** ICGFI	**20**

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